A Multivariate Linear Model for Environmental Pollutant Dispersion in New York City(Proposal)

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Abstract: Problem and Objectives

The precise prediction of multi-pollutant dispersion is crucial for urban air quality management and public health advocacy. We focus on two primary pollutants, Nitrogen Oxides (NO_x), Fine Particulate Matter ($PM_{2.5}$), and Sulfur Dioxide (SO_2). Also, the settings of dispersion such as urban corridors and industrial zones. Modeling these pollutants is significant to observe the scale of their impact on air quality in relation to the settings of dispersion. Our variables being meteorological factors as predictors such as wind, humidity and topography. A multivariate approach is necessary due to the gaps that exist in traditional univariate models as they fail to consider the special correlations and cross-pollutant interactions fundamental to complex environmental systems. The model will follow the form $Y= XB+E$ (or $Y = X\beta+\varepsilon$), where Y is the response matrix consisting of the pollutants, X is the multi-meteorological predictor matrix, B is the coefficient (line of best fit for the regression model), and E is the error (or residual) of the model. This approach delivers a more holistic comprehension of the chemical and physical dynamics governing modern urban environments.

Identifying data sets

Pollutant data (Y Matrix): Hourly concentration data for NO_x, $PM_{2.5}$, and SO_2

- *Source:* EPA Air Quality System (AQS), OpenAQ, or a local Environmental Protection Bureau.

Meteorological Data (X Matrix): Wind speed, wind direction, relative humidity, and ambient temperature.

- *Source:* NOAA National Centers for Environmental Information (NCEI) or local airport weather stations.

Topographical Data: Digital Elevation Models (DEM) to account for terrain effects in industrial zones.

- *Source:* USGS EarthExplorer

Software and Analytical Tools

The analysis requires tools capable of matrix algebra and statistical validation:

- **Programming Language: Python** (using pandas for data manipulation, scikit-learn for the multivariate regression, and *statsmodels* for rigorous statistical testing) or **R** (using the lm () or mlm functions).
- **Geospatial Analysis: QGIS** or **ArcGIS** to map the dispersion patterns and extract topographical variables for the X matrix.
- **Visualization: Matplotlib/Seaborn** (Python) or **ggplot2** (R) to create correlation heatmaps and residual plots (E).

Expected Day-to-Day Activities

This project will be executed in four phases:

Phase 1: Data Acquisition & Cleaning (Weeks 1-2)

- Daily Activity: Downloading raw CSV/API data from environmental sensors.
- Handling missing values, synchronizing time-stamps between weather data and pollutant sensors, and normalizing data scales.

Phase 2: Exploratory Data Analysis (EDA) (Weeks 3-4)

- Daily Activity: Generating Pearson correlation matrices to identify initial relationships between pollutants.
- Visualizing "pollution roses" to see how wind direction affects concentration levels in industrial zones.

Phase 3: Model Development & Training (Weeks 5-7)

- Daily Activity: Constructing the Y and X matrices.
- Running the multivariate linear regression and performing "Leave-One-Out" Cross-Validation (LOOCV) to ensure the model isn't overfitting.
- Analyzing the B (coefficient) matrix to determine which meteorological factor has the highest impact.

Phase 4: Validation & Thesis Writing (Weeks 8-10)

- Daily Activity: Evaluating the E (error) term for heteroscedasticity (checking if error is consistent).
- Comparing results against a standard univariate baseline to quantify the improvement in accuracy.
- Finalizing the research report and visualizations.